CYD_Panel (Cheap Yellow Display)

Quick Reference

Arduino (IDE) Experiment Control Panel

Jaap Daniëlse V1.2 - 30-04-2026 – Panel V1.x.x (Dual Core)

Panel Functions

```
Panel.begin();
Panel.receive();
Panel.send(channel); //void
Panel.send(channel, bool);
Panel.send(channel, uint8_t / byte);
Panel.send(channel, int/int16_t/int32_t);
Panel.send(channel, const char*);
Panel.send(channel, param);
Panel.send(channel, param, int / int16_t / int32_t);
Panel.send(channel, param, const char*);
Panel.send(channel, param, color[2]);
Panel.send(channel, param, font[3]);
Panel.send(channel, layout[1]);
Panel.sendf(channel, const char*, ...);
```

Minimal code

```
#include "CYD_panel.h"
void setup() {
  Panel.begin();
}

void loop() {
  Panel.receive();
  ... // user code
}

void PannelCallback(CYD_channel event) {
  switch(event) {
    case channel: // for each ch. needed
      ... // user code
      break;
  }
}
```

Channels / events

channel: CYD_ApplicationName

chan: CYD_MonitorField_'x' (1..6) send(f)

const char*	display text
CYD_COLOR, color[2]	text color
CYD_BACKCOLOR,col.	backgr. color
CYD_FONT, font[3]	text font
bool	visible / invisible

channel: CYD_Display_'x' (1..6)

channel: CYD_Button_'x' (1..12) .send(f)

const char*	display text
CYD_COLOR, color[2]	text color
CYD_BACKCOLOR,col[2]	backgr. color
CYD_FONT, font[3]	text font
bool	visible / invisible

.receive

void	object click

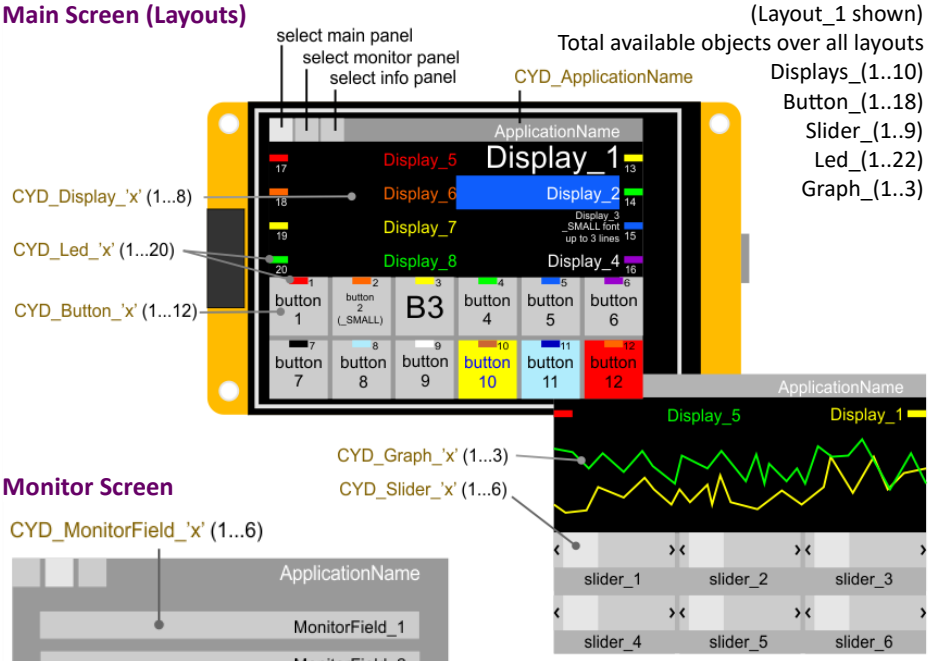
channel: CYD_Led_'x' (1..18) .send

CYD_COLOR, color[2]	LED color
bool	visible

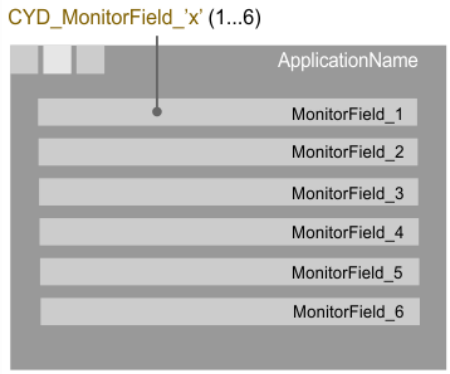
channel: CYD_Connected .receive

void	panel connected

Main Screen (Layouts)



Monitor Screen



channel: CYD_Slider_'x' (1..6) .send(f)

const char*	display text
int[4]	initial value
_MaxValue, int[4]	max sl. value
CYD_COLOR, color[2]	text color
CYD_BACKCOLOR, col.	backgr. color
CYD_FONT, font[3]	text font
bool	visible / inv.

.receive

int	slider value

channel: CYD_Graph_'x' (1..3) .send

char (int8_t)	value
CYD_VALUECOUNT,int[4]	value count
CYD_ROLLING (def.)	sel. rolling gr.
CYD_STATIC	sel. static gr.
CYD_CLEAR	clear gr. values
CYD_COLOR, color[2]	gr. line color
bool	visible/inv.

channel: CYD_InfoPanel .send

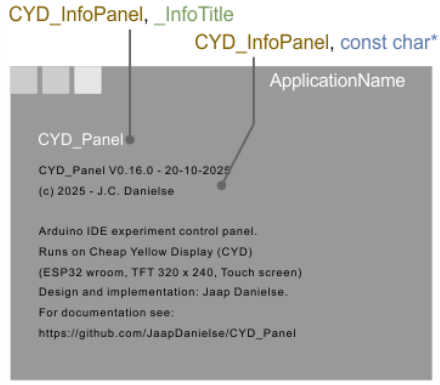
_InfoTitle, const char *	title text
const char*	text line (10)

! use .send() only (no sendf())

channel: CYD_Reset .send

void	reset to current layout
layout[1]	reset to layout

InfoPanel



Panel variables

(Event data received)

cyd_types

cyd_void	void
cyd_int	int

Panel variable type

Panel.cydr_type	cyd_type

Panel variables

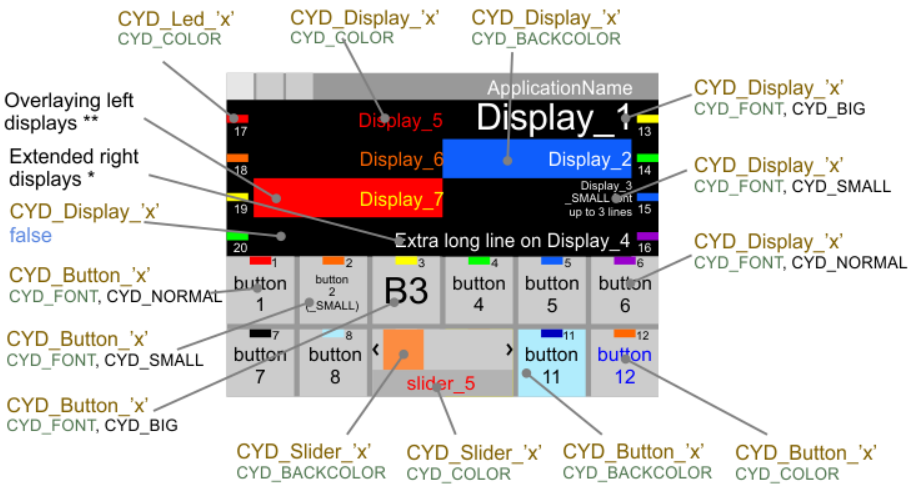
Panel.cydr_void	void
Panel.cydr_int	int (int16_t)

Example:

```
if(Panel.cydr_type == CYD_int)
  myInt = Panel.cydr_int
```

[1] see: Layouts
[2] see: Colors
[3] see: Fonts
[4] int, int16_t, int32_t

Channel Attributes



Overlapping displays
* Right side (text) displays extend underneath the left side displays.
** Left side (text) displays overlap right side displays when visible.
Overlaying displays are redrawn automatically.
Write (text) in right display is not truncated.

Examples:
Panel.send(CYD_Led_17, _Color, _RED);
Panel.send(CYD_Slider_1, _BackColor, _BLUE);
Panel.send(CYD_Button_12, _Color, _LBLUE);
Panel.send(CYD_Display_1, _Font, _BIG);
Panel.send(CYD_Display_8, false); // not visible

Colors

Channels: CYD_ApplicationName, CYD_Display, CYD_Button, CYD_Led, CYD_Slider, CYD_Graph + Param: CYD_COLOR, CYD_BACKCOLOR + color

CYD_BLACK	
CYD _GRAY	
CYD _PURPLE	
CYD _PINK	
CYD _LBLUE	
CYD _BLUE	
CYD _DBLUE	
CYD _GREEN	
CYD _YELLOW	
CYD _ORANGE	
CYD _RED	
CYD _BROWN	
CYD _WHITE	

Example:
Panel.send(Button_1, _Color, _RED);

Layouts

Function Panel.begin()
Channels: CYD_Reset

CYD_LAYOUT_1*(def.)*	see layout 1 diagr.
CYD_LAYOUT_2	see layout 2 diagr.
CYD_LAYOUT_3	see layout 3 diagr.
CYD_LAYOUT_4	see layout 4 diagr.

Examples:
Panel.begin(CYD_LAYOUT_2);
Panel.send(CYD_Reset); // current layout
Panel.send(CYD_Reset, CYD_LAYOUT_2);

Fonts

Channels: CYD_ApplicationName, CYD_Display, CYD_Button, CYD_Led, CYD_Slider, CYD_Graph + Param: CYD_FONT + font

CYD_SMALL	small font (8px)
CYD_NORMAL(def.)	normal font (16px)
CYD_BIG	big font (28px)

Example:
Panel.send(CYD_Button_1, CYD_FONT, CYD_SMALL);

Sendf() / Printf formatting

%[flags][width][length]specifier

specifiers (limited list)

%c	ascii char	byte
%d	signed dec.	int16
%ld	signed dec.	int32
%u	unsigned dec.	uint16
%lu	unsigned dec.	uint32
%o	unsigned octal	any
%x	uns. hex lc/uc	any
%s	string	char*
%f*	float	float

*Not ATMega (UNO R3, Nano)

flags

-	left justify- in padding
+	force sign
0	pad zero's
x.y	float chars(x).decimals(y)

Examples:
Panel.sendf (CYD_Display_1, "Test %d", 10);
// output: Test 10
Panel.sendf(CYD_Display_1, "Test %03d", 10);
// output: Test 010
Panel.sendf(CYD_Display_1, "Test %+d", 10);
// output: Test +10
Panel.sendf(CYD_Display_1, "Test %f", 10.123);
// output: Test 10.123000
Panel.sendf(CYD_Display_1, "T %.2f", 10.123);
// output: Test 10.12
Panel.sendf(CYD_Display_1, "T %6.2f", 10.123);
// output: Test 010.12

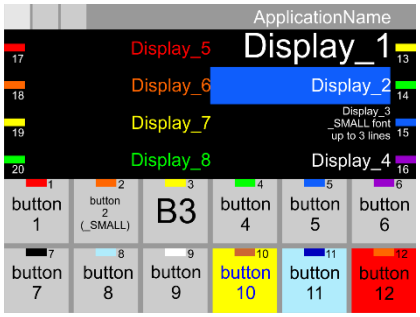
Screen Layouts

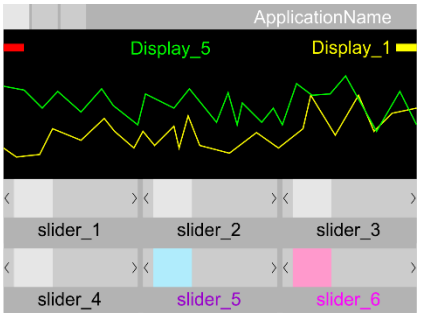
Four available layouts: _LAYOUT_(1..4)
(*see below*)

Select layout:
Panel.begin(); // def. _LAYOUT_1
Panel.begin(CYD_LAYOUT_x);
Panel.send(CYD_reset); // def. LAYOUT 1
Panel.send(CYD_reset, CYD_LAYOUT_x);
(*see example sketches: F_MultiPanelMultiLayout.ino*)

Total available objects over all layouts
CYD_Displays_(1..10)
left displays overlay right
CYD_Button_(1..18)
CYD_Slider_(1..9)
Sliders overlap 2 buttons
CYD_Led_(1..22)
CYD_Graph_(1..3)

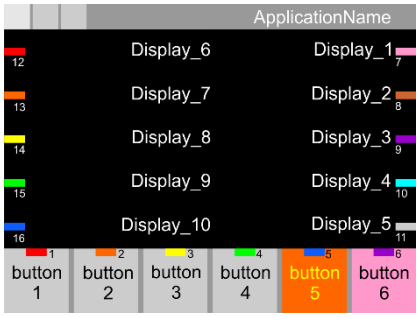
LAYOUT_1

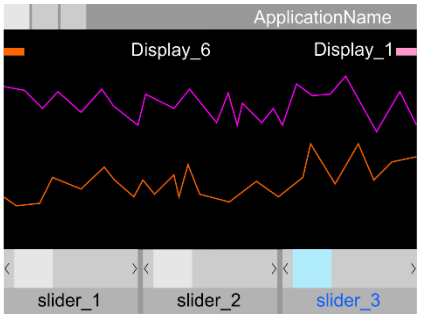




Display panel **320 x 116**
CYD_Display_(1..8)
Display (text): **280 x 29 (x5) (1..4)**
Display (text): **280 x 29 (x5) (5..8)**
- small font 3 lines
- normal font 1 line
- large font 1 line
CYD_Button_(1..12)
CYD_Led_(1..20)
CYD_Graph_(1...3)

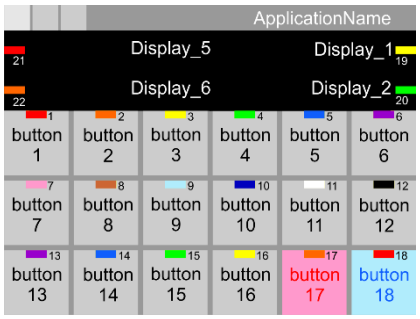
LAYOUT_2

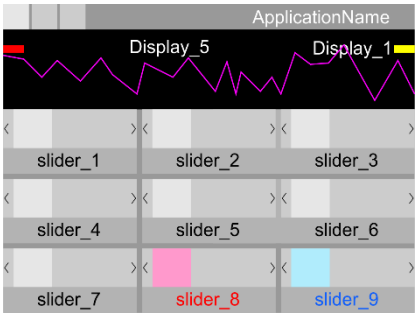




Display panel **320 x 170**
CYD_Display_(1..10)
Display (text): **280 x 34 x5 (1..5)**
Display (text): **140 x 34 x5 (6..10)**
- small font 3 lines
- normal font 2 lines
- large font 1 line
CYD_Button_(1..6)
CYD_Led_(1..16)
CYD_Graph_(1...3)

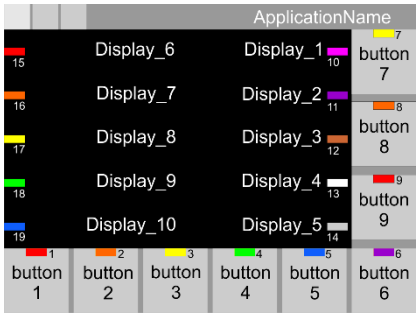
LAYOUT_3

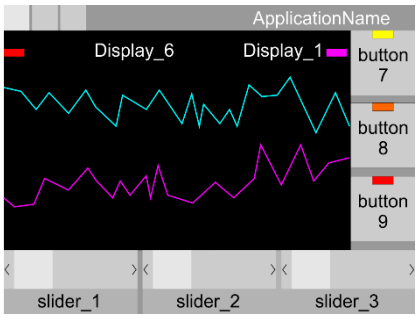




Display panel **320 x 62**
CYD_Display_(1..4)
Display (text): **280 x 31 x2 (1..2)**
Display (text): **140 x 31 x2 (3..4)**
- small font 3 lines
- normal font 1 line
- large font 1 line
CYD_Button_(1..18)
CYD_Led_(1..22)
CYD_Graph_(1...3)

LAYOUT_4





Display panel **320 x 170**
CYD_Display_(1..10)
Display (text): **226 x 34 x5 (1..5)**
Display (text): **113 x 34 x5 (6..10)**
- small font 3 lines
- normal font 2 lines
- large font 1 line
CYD_Button_(1..9)
CYD_Led_(1..19)
CYD_Graph_(1...3)

Basic Example Code

```
#include "CYD_Panel.h" // includes the CYD_Panel libray

bool Switch = false; // Global Boolean (true/false) variable

void setup() {
  Panel.begin(); // Start the CYD_Panel
}

void loop() {
  Panel.receive(); // added in the loop to catch panel events (e.g. Button pressed)
}

void PanelCallback(CYD_channel event) { // panel events are sent here

  switch(event) { // the switch is used to detect what has happend
    case CYD_PanelConnected: // the pannel is connected, we can configure it
      Panel.send(CYD_Button_1, "on/off"); // write “on/off” to button_1 and make it visible
      break; // no more config needed for now

    case CYD_Button_1: // Button_1 has been pressed
      Switch = !Switch; // set the switch variable to what it was not (true>false / false>true)
      if(Switch) // if switch is true
        Panel.send(CYD_Display_1, "switch on"); // send “switch on” to Dispay_1
      else // if switch is false
        Panel.send(CYD_Display_1, "switch off"); // send “switch off” to Dispay_1
      break;  // that is all for the Button_1 pressed action

    default: break; // if anything else comes in from the panel, do nothing
  }
}
```

CYD_Panel_ESPNow

Jaap Daniëlse V1.1 - 30-04-2026 – CYD_Panel_ESPNow V1.x.x / CYD_Panel V1.0.0

CYD_Panel as remote via ESP NOW

Use the Cheap Yellow Display Panel as a remote control and display for any ESP32 (including Arduino Nano ESP32) or ESP8266

On the Cheap Yellow Display Panel

Load “ESPNow-Connection.ino” from the CYD_Panel_ESPNow library examples into the Cheap Yellow Display.

Modify Wifi Channel

The Wifi Channel may be set by modifying the line:
const int WiFiChannel = 1;
Available channel numbers are 1 through 14. Where 14 is not allowed in all countries.
When modified the same channel must be set on the client side.

Debugging

The Arduino IDE console will show ESPNow connection info and errors when connected. (set speed to 115200).

Panel Client

In your client sketch, instead of the CYD_Panel library, include the CYD_Panel_ESPNow library:

#include “CYD_Panel_ESPNow.h”
(*see CYD_Panel_ESPNow example sketches*)

Modify Wifi Channel

The wifi channel may be modified by inserting (before Panel.begin();):

WifiChannel = x;

Where x is the desired WifiChannel (1-14).

Debugging

Adding in setup();
Serial.begin(115200);
Will show ESPNow connection info and errors on the Arduino IDE console.

Connection behavior

When the CYD_Panel is switched on it waits for a connection on the channel set.

When the client is started sends a pairing request.

If the Panel receives a pairing request with the right version id. It responds with a pairing request back to the client.
The Panel and client are now paired.
The panel will only connect with this client until it is reset.

The paired client may be reset and will re-connect.